REQUIRED DELIVERABLE

Explainability Layer

How COMPASSIQ explains maturity diagnosis, scores, blockers, anomalies and RAG roadmap recommendations.

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Prepared for GitHub repository submission

1. Explainability Objective

The explainability layer ensures that every COMPASSIQ output is traceable. Users and judges can understand why a maturity stage was assigned, how each score was computed, which evidence contributed to the result, which blockers were detected, and which knowledge base resource supports each roadmap action.

A recommendation without evidence, a score without sub-criteria, or a roadmap action without a source is treated as incomplete output.

2. Explainability Principles

Principle	Implementation in COMPASSIQ
Traceability	Each output links back to project profile fields, questionnaire answers, score criteria or knowledge base IDs.
Decomposition	Composite scores are decomposed into sub-scores and criterion-level contributions.
Plain language	Users receive short non-technical explanations and improvement actions.
Uncertainty	Incomplete or contradictory profiles surface uncertainty instead of hiding it.
Grounding	Roadmap resources cite the specific KB item, URL and category used by retrieval.

3. Diagnostic Explainability

The diagnostic engine assigns one maturity stage from a six-stage taxonomy: Ideation, Market Validation, Structuration, Fundraising, Launch Planning and Growth. The explanation shows evidence that supports the stage, missing evidence that blocks progression, and any perception-reality divergence.

Diagnostic output	Explanation surfaced to the user
Assigned maturity stage	The stage is shown with the strongest supporting signals from the project profile.
Confidence level	Confidence is reduced when answers are missing, inconsistent or weakly evidenced.
Perception-reality gap	If the founder self-assesses at a higher/lower stage, the system explains the exact missing criteria.
Priority blockers	Blockers are ranked by severity and mapped to domains: market, financial, legal, technical or organizational.
Next evidence requested	The system asks for the highest-value missing evidence required to refine the diagnosis.

4. Scoring Methodology and Weights

Each composite score is computed from sub-criteria with documented weights. Scores are not presented as black-box labels; each score card exposes sub-score values, weights, evidence and improvement guidance.

Composite score	Sub-criteria and weights
Market Score	Customer validation evidence 35%; revenue model clarity 25%; addressable market and competition 25%; early traction 15%.
Commercial Offer Score	Value proposition clarity 25%; differentiation 25%; product/service readiness 25%; pricing and customer-fit coherence 25%.
Innovation Score	Local novelty 30%; technology intensity 25%; barrier to entry 25%; departure from existing offerings 20%.

Composite score	Sub-criteria and weights
Scalability Score	Replicability without linear cost 30%; manual dependency 25%; deployment cost structure 25%; geographic/market expansion potential 20%.
Green Score	Environmental impact 30%; SDG alignment 25%; resource efficiency 25%; circular economy / footprint mitigation 20%.

5. Aggregation Logic

Each sub-criterion is normalized on a 0–100 scale. The weighted composite score is calculated as the sum of normalized sub-score × weight. However, critical blockers can cap the final score to prevent a strong secondary criterion from hiding a fundamental weakness.

Rule	Purpose	Example
Weighted average	Combines sub-criteria according to documented importance.	Market Score = validation × 35% + revenue clarity × 25% + market/competition × 25% + traction × 15%.
Critical cap	Prevents misleading high scores when a must-have signal is absent.	If customer validation = 0, Market Score cannot exceed 60 even if market size is large.
Uncertainty penalty	Reflects incomplete evidence.	Missing pricing data lowers Commercial Offer confidence and explanation asks for pricing assumptions.
Anomaly flag	Surfaces contradictory claims.	High claimed traction with no customers or revenue evidence is flagged.

6. RAG and Roadmap Explainability

The roadmap generator uses the diagnostic stage, low sub-scores, blockers, sector and geography as retrieval context. Every roadmap action must cite one or more knowledge base items by ID.

Trigger	Retrieved resource example	Explanation
AgriTech validation gap	kb_024 Cap Bon AgriTech Hub; kb_028 Mobidoo	These resources match early validation and mentoring needs for an AgriTech founder.
Agricultural structuration need	kb_025 APIA; kb_036 CTAB	These resources match agricultural investment support and organic/agri standards.
Financing readiness gap	kb_002 BFPME; kb_032 Green Tunisia Fund	Financing resources are recommended only after validation evidence is improved.
Administrative/legal setup	kb_003 Startup Act; kb_013 Guichet Unique; kb_014 CNSS	Administrative resources are triggered by legal form and launch planning needs.

7. Example Explanation Card — Amina, AgriTech Founder

Output	Explanation shown
Self-assessed stage	Fundraising-ready
System diagnosis	Structuration / Market Validation gap
Why	No validated customer evidence; revenue model unclear; scalability depends on manual accompaniment.
Market Score	62/100 — customer validation is the main weakness despite a relevant market pain.
Scalability Score	54/100 — deployment is still highly manual and not yet replicable at low marginal cost.
Recommended next action	Run farmer interviews and pilot validation before preparing financing file.

Output	Explanation shown
Sources	kb_024 Cap Bon AgriTech Hub; kb_025 APIA; kb_036 CTAB; kb_032 Green Tunisia Fund; kb_002 BFPME.

8. UI Components of the Explainability Layer

• Diagnostic card: assigned stage, confidence, evidence used, missing evidence and perception-gap warning.

• Score breakdown modal: composite score, sub-scores, weights, criterion contributions and improvement action.

• Anomaly banner: contradictory claims and missing proof are shown as warnings, not hidden.

• Roadmap action card: action, time horizon, rationale, linked score/blocker and cited KB source ID.

• Mon Parcours view: tracks past recommendations, completed actions, score evolution and updated next steps.

9. GitHub Implementation Notes

Recommended repository files: scoring_methodology.md for weights and caps, evaluation_report.pdf for metrics, knowledge_base_documentation.pdf for sources, and explainability_layer.pdf for UI/logic documentation. Each output object should include an explanation field and a source_ids array when using RAG.